Membership constraints

CLEANING DATA IN PYTHON

Adel Nehme
Content Developer @DataCamp

datacamp

Chapter 2 - Text and categorical data problems

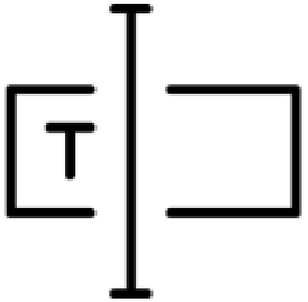
Categories and membership constraints

Predefined finite set of categories

Type of data	Example values	Numeric representation
Marriage Status	`unmarried`, `married`	`0`,`1`
Household Income Category	`0-20K`, `20-40K`, ...	`0`,`1`,..
Loan Status	`default`,`payed`,`no_loan`	`0`,`1`,`2`

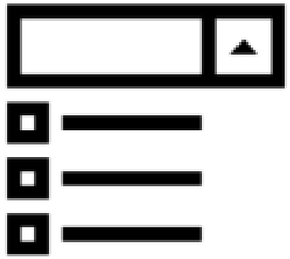
*Marriage status can **only** be* `unmarried` _or_ `married`

Why could we have these problems?



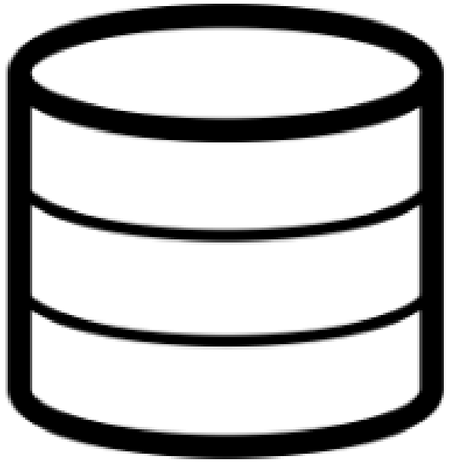
Free text

Or



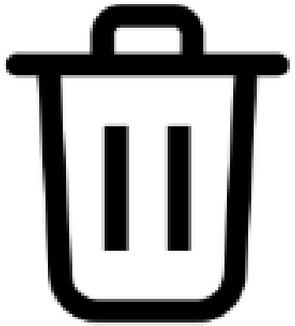
Dropdowns

Data Entry Errors

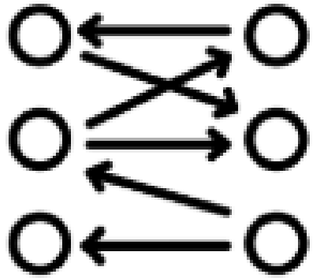


Parsing Errors

How do we treat these problems?



Dropping Data



Remapping Categories

Inferring Categories

An example

```
# Read study data and print it
study_data = pd.read_csv('study.csv')
study_data
```

```
      name   birthday blood_type
1     Beth 2019-10-20         B-
2 Ignatius 2020-07-08         A-
3     Paul 2019-08-12         O+
4    Helen 2019-03-17         O-
5 Jennifer 2019-12-17         Z+
6  Kennedy 2020-04-27         A+
7    Keith 2019-04-19        AB+
```

```
# Correct possible blood types
categories
```

```
  blood_type
1         O-
2         O+
3         A-
4         A+
5         B+
6         B-
7        AB+
8        AB-
```

An example

```
# Read study data and print it
study_data = pd.read_csv('study.csv')
study_data
```

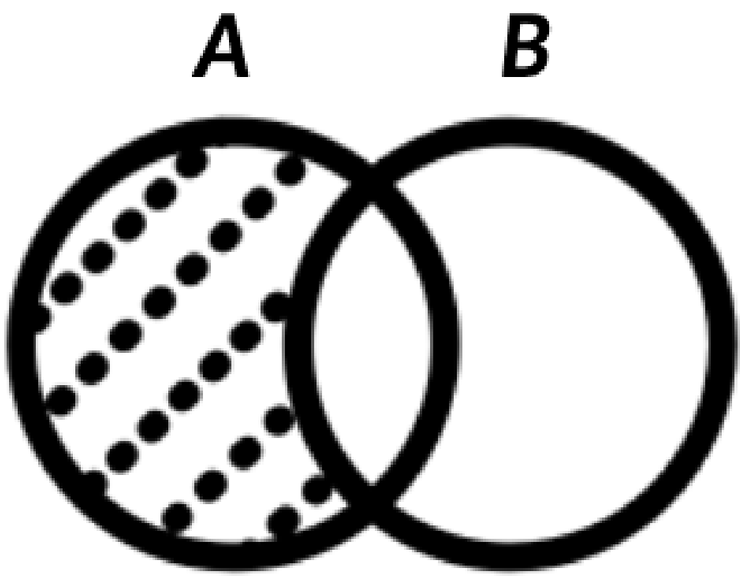
```
      name   birthday blood_type
1     Beth 2019-10-20         B-
2 Ignatius 2020-07-08         A-
3     Paul 2019-08-12         O+
4    Helen 2019-03-17         O-
5 Jennifer 2019-12-17         Z+  <--
6  Kennedy 2020-04-27         A+
7    Keith 2019-04-19        AB+
```

```
# Correct possible blood types
categories
```

```
  blood_type
1         O-
2         O+
3         A-
4         A+
5         B+
6         B-
7        AB+
8        AB-
```

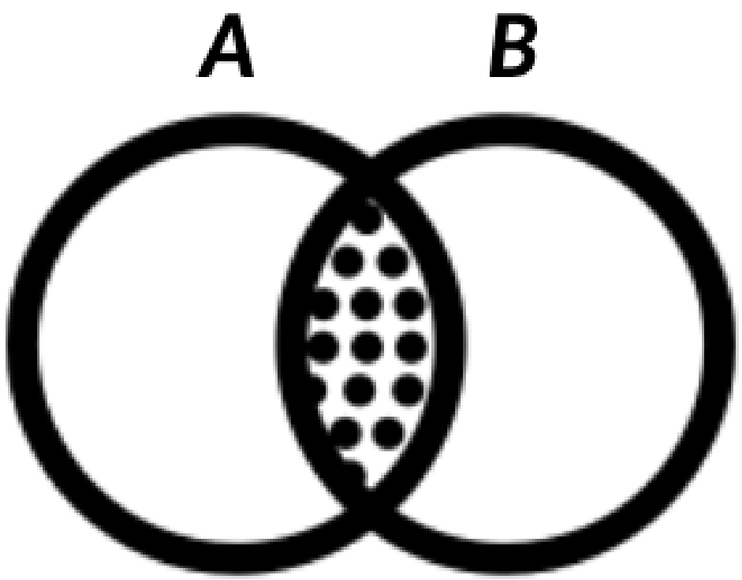
A note on joins

Anti Joins



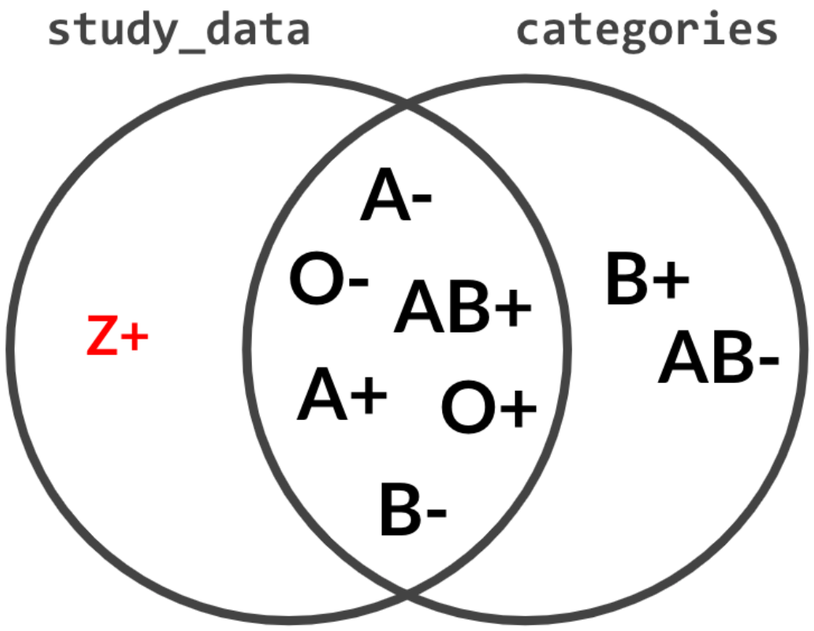
What is in A and not in B

Inner Joins



What is in both A and B

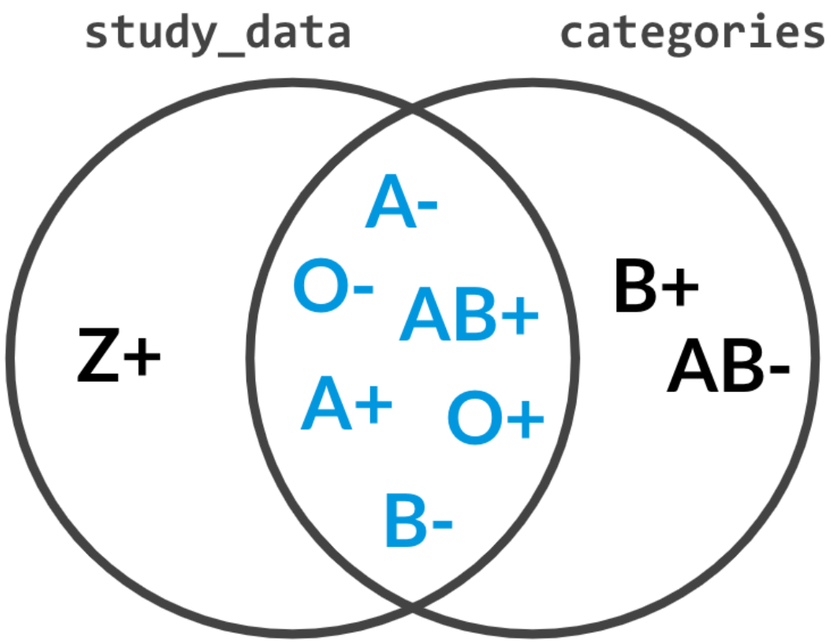
A left anti join on blood types



What is in `study_data` *only*

Returns only rows containing Z+

An inner join on blood types



What is in `study_data` *and* `categories` *only*

Returns all the rows except those containing Z+ , B+ *and* AB-

Finding inconsistent categories

```
inconsistent_categories = set(study_data['blood_type']).difference(categories['blood_type'])
print(inconsistent_categories)
```

```
{'Z+'}
```

```
# Get and print rows with inconsistent categories
inconsistent_rows = study_data['blood_type'].isin(inconsistent_categories)
study_data[inconsistent_rows]
```

```
      name   birthday blood_type
5 Jennifer 2019-12-17         Z+
```

Dropping inconsistent categories

```
inconsistent_categories = set(study_data['blood_type']).difference(categories['blood_type'])
inconsistent_rows = study_data['blood_type'].isin(inconsistent_categories)
inconsistent_data = study_data[inconsistent_rows]
# Drop inconsistent categories and get consistent data only
consistent_data = study_data[~inconsistent_rows]
```

```
      name   birthday blood_type
1     Beth 2019-10-20         B-
2 Ignatius 2020-07-08         A-
3     Paul 2019-08-12         O+
4    Helen 2019-03-17         O-
...    ...        ...        ...
```

Let's practice!

CLEANING DATA IN PYTHON

Categorical variables

CLEANING DATA IN PYTHON

Adel Nehme
Content Developer @DataCamp

datacamp

What type of errors could we have?

I) Value inconsistency

- *Inconsistent fields:* `'married'`, `'Maried'`, `'UNMARRIED'`, `'not married'`..
- _Trailing white spaces: _ `'married '`, `' married '`..

II) Collapsing too many categories to few

- *Creating new groups:* `0-20K`, `20-40K` categories ... from continuous household income data
- *Mapping groups to new ones:* Mapping household income categories to 2 `'rich'`, `'poor'`

III) Making sure data is of type `category` (seen in Chapter 1)

Value consistency

Capitalization: `'married'`, `'Married'`, `'UNMARRIED'`, `'unmarried'`..

```
# Get marriage status column
marriage_status = demographics['marriage_status']
marriage_status.value_counts()
```

```
unmarried    352
married      268
MARRIED      204
UNMARRIED    176
dtype: int64
```

Value consistency

```
# Get value counts on DataFrame
marriage_status.groupby('marriage_status').count()
```

```
                household_income  gender
marriage_status
MARRIED                      204     204
UNMARRIED                    176     176
married                      268     268
unmarried                    352     352
```

Value consistency

```
# Capitalize
marriage_status['marriage_status'] = marriage_status['marriage_status'].str.upper()
marriage_status['marriage_status'].value_counts()
```

```
UNMARRIED    528
MARRIED      472
```

```
# Lowercase
marriage_status['marriage_status'] = marriage_status['marriage_status'].str.lower()
marriage_status['marriage_status'].value_counts()
```

```
unmarried    528
married      472
```

Value consistency

Trailing spaces: `'married '`, `'married'`, `'unmarried'`, `' unmarried'`..

```
# Get marriage status column
marriage_status = demographics['marriage_status']
marriage_status.value_counts()
```

```
 unmarried   352
unmarried    268
married      204
married      176
dtype: int64
```

Value consistency

```
# Strip all spaces
demographics = demographics['marriage_status'].str.strip()
demographics['marriage_status'].value_counts()
```

```
unmarried    528
married      472
```

Collapsing data into categories

Create categories out of data: `income_group` column from `income` column.

```
# Using qcut()
import pandas as pd
group_names = ['0-200K', '200K-500K', '500K+']
demographics['income_group'] = pd.qcut(demographics['household_income'], q = 3,
                                       labels = group_names)
# Print income_group column
demographics[['income_group', 'household_income']]
```

```
     category  household_income
0   200K-500K  189243
1       500K+  778533
..
```

Collapsing data into categories

Create categories out of data: `income_group` column from `income` column.

```
# Using cut() - create category ranges and names
ranges = [0,200000,500000,np.inf]
group_names = ['0-200K', '200K-500K', '500K+']
# Create income group column
demographics['income_group'] = pd.cut(demographics['household_income'], bins=ranges,
                                      labels=group_names)
demographics[['income_group', 'household_income']]
```

```
     category  Income
0      0-200K  189243
1       500K+  778533
```

Collapsing data into categories

Map categories to fewer ones: reducing categories in categorical column.

`operating_system` column is: `'Microsoft', 'MacOS', 'IOS', 'Android', 'Linux'`

`operating_system` column should become: `'DesktopOS', 'MobileOS'`

```
# Create mapping dictionary and replace
mapping = {'Microsoft':'DesktopOS', 'MacOS':'DesktopOS', 'Linux':'DesktopOS',
           'IOS':'MobileOS', 'Android':'MobileOS'}
devices['operating_system'] = devices['operating_system'].replace(mapping)
devices['operating_system'].unique()
```

```
array(['DesktopOS', 'MobileOS'], dtype=object)
```

Let's practice!

CLEANING DATA IN PYTHON

Cleaning text data

CLEANING DATA IN PYTHON

Adel Nehme
VP of AI Curriculum, DataCamp

datacamp

What is text data?

Type of data	Example values
Names	`Alex`, `Sara` ...
Phone numbers	`+96171679912` ...
Emails	`adel@datacamp.com`..
Passwords	...

Common text data problems

1) Data inconsistency:

`+96171679912` or `0096171679912` or ..?

2) Fixed length violations:

Passwords needs to be at least 8 characters

3) Typos:

`+961.71.679912`

Example

```
phones = pd.read_csv('phones.csv')
print(phones)
```

```
              Full name      Phone number
0       Noelani A. Gray  001-702-397-5143
1        Myles Z. Gomez  001-329-485-0540
2          Gil B. Silva  001-195-492-2338
3    Prescott D. Hardin   +1-297-996-4904
4    Benedict G. Valdez  001-969-820-3536
5      Reece M. Andrews              4138
6        Hayfa E. Keith  001-536-175-8444
7       Hedley I. Logan  001-681-552-1823
8      Jack W. Carrillo  001-910-323-5265
9       Lionel M. Davis  001-143-119-9210
```

Example

```python
phones = pd.read_csv('phones.csv')
print(phones)
```

```
              Full name      Phone number
0       Noelani A. Gray  001-702-397-5143
1        Myles Z. Gomez  001-329-485-0540
2          Gil B. Silva  001-195-492-2338
3    Prescott D. Hardin   +1-297-996-4904   <-- Inconsistent data format
4    Benedict G. Valdez  001-969-820-3536
5      Reece M. Andrews              4138   <-- Length violation
6        Hayfa E. Keith  001-536-175-8444
7       Hedley I. Logan  001-681-552-1823
8      Jack W. Carrillo  001-910-323-5265
9       Lionel M. Davis  001-143-119-9210
```

Example

```
phones = pd.read_csv('phones.csv')
print(phones)
```

```
              Full name   Phone number
0       Noelani A. Gray  0017023975143
1        Myles Z. Gomez  0013294850540
2          Gil B. Silva  0011954922338
3    Prescott D. Hardin  0012979964904
4    Benedict G. Valdez  0019698203536
5      Reece M. Andrews           NaN
6        Hayfa E. Keith  0015361758444
7       Hedley I. Logan  0016815521823
8      Jack W. Carrillo  0019103235265
9       Lionel M. Davis  0011431199210
```

Fixing the phone number column

```
# Replace "+" with "00"
phones["Phone number"] = phones["Phone number"].str.replace("+", "00")
phones
```

```
            Full name      Phone number
0     Noelani A. Gray  001-702-397-5143
1      Myles Z. Gomez  001-329-485-0540
2        Gil B. Silva  001-195-492-2338
3  Prescott D. Hardin  001-297-996-4904
4  Benedict G. Valdez  001-969-820-3536
5    Reece M. Andrews              4138
6      Hayfa E. Keith  001-536-175-8444
7     Hedley I. Logan  001-681-552-1823
8    Jack W. Carrillo  001-910-323-5265
9     Lionel M. Davis  001-143-119-9210
```

Fixing the phone number column

```
# Replace "-" with nothing
phones["Phone number"] = phones["Phone number"].str.replace("-", "")
phones
```

```
             Full name   Phone number
0     Noelani A. Gray  0017023975143
1      Myles Z. Gomez  0013294850540
2        Gil B. Silva  0011954922338
3  Prescott D. Hardin  0012979964904
4  Benedict G. Valdez  0019698203536
5    Reece M. Andrews           4138
6      Hayfa E. Keith  0015361758444
7     Hedley I. Logan  0016815521823
8    Jack W. Carrillo  0019103235265
9     Lionel M. Davis  0011431199210
```

Fixing the phone number column

```
# Replace phone numbers with lower than 10 digits to NaN
digits = phones['Phone number'].str.len()
phones.loc[digits < 10, "Phone number"] = np.nan
phones
```

```
               Full name   Phone number
0        Noelani A. Gray  0017023975143
1         Myles Z. Gomez  0013294850540
2           Gil B. Silva  0011954922338
3     Prescott D. Hardin  0012979964904
4     Benedict G. Valdez  0019698203536
5       Reece M. Andrews           NaN
6         Hayfa E. Keith  0015361758444
7        Hedley I. Logan  0016815521823
8       Jack W. Carrillo  0019103235265
9        Lionel M. Davis  0011431199210
```

Fixing the phone number column

```
# Find length of each row in Phone number column
sanity_check = phone['Phone number'].str.len()
```

```
# Assert minmum phone number length is 10
assert sanity_check.min() >= 10
```

```
# Assert all numbers do not have "+" or "-"
assert phone['Phone number'].str.contains("+|-").any() == False
```

Remember, `assert` ***returns nothing if the condition passes***

But what about more complicated examples?

```
phones.head()
```

```
          Full name    Phone number
0     Olga Robinson  +(01706)-25891
1       Justina Kim    +0500-571437
2    Tamekah Henson      +0800-1111
3     Miranda Solis   +07058-879063
4  Caldwell Gilliam  +(016977)-8424
```

Supercharged control + F

Regular expressions in action

```
# Replace letters with nothing
phones['Phone number'] = phones['Phone number'].str.replace(r'\D+', '')
phones.head()
```

```
          Full name Phone number
0     Olga Robinson   0170625891
1       Justina Kim   0500571437
2    Tamekah Henson     08001111
3     Miranda Solis  07058879063
4  Caldwell Gilliam   0169778424
```

Let's practice!

CLEANING DATA IN PYTHON